

ITS Scorecard Technical Report

Executive Summary

This technical report presents the development and implementation of a comprehensive evaluation framework for the deployment of Intelligent Transportation Systems (ITS). The objective of this project is to establish a standardized, data-driven methodology that quantitatively assesses the performance, institutional readiness, and future development trajectory of ITS programs across transportation agencies.

The proposed framework organizes the evaluation into five primary dimensions: incremental ITS deployment benefit–cost change, ITS coverage within existing infrastructure, policy and legislative readiness, integration of ITS within funded and planned transportation programs, and institutional capacity represented by ITS facilities and operational support infrastructure. Each dimension is supported by clearly defined metrics, normalized scoring methods, and unified rating procedures to ensure consistent comparison across jurisdictions and over time.

To operationalize the framework, the project integrates multiple data sources, including ITS cost and benefit databases, deployment survey records, policy and legislative documentation, transportation planning programs, and institutional infrastructure information related to ITS operations and testing environments. These datasets are processed to produce standardized component scores and an aggregated annual ITS Scorecard rating that reflects both current deployment conditions and forward-looking ITS development readiness.

In addition to establishing the evaluation methodology, the project implements a web-based ITS Scorecard application that supports interactive dashboard visualization, deployment analytics, project tracking, and development forecasting. The platform enables centralized data management, periodic updates, and transparent performance reporting, providing transportation agencies and stakeholders with a structured tool to monitor ITS progress and support data-driven decision-making for future ITS investments.

1. Evaluation Criteria

This section defines the structured criteria used to evaluate ITS projects at the annual scale. The evaluation framework is organized around four core dimensions: economic performance, infrastructure coverage, legislative progress, and integration within planned transportation improvements. Each criterion is defined in a measurable, standardized manner to enable cross-jurisdictional comparisons and longitudinal tracking.

1.1 ITS Deployment Benefit/Cost (B/C) Analysis

The economic performance of ITS deployment is evaluated using an annual incremental benefit–cost framework. Unlike conventional project appraisal methods that assess a single lifecycle benefit–cost ratio at project completion, this criterion evaluates year-over-year changes in monetized benefits relative to annualized deployment and operational costs. The objective is to measure sustained economic effectiveness and performance progression rather than static, one-time project valuation.

The evaluation incorporates three benefit domains:

- Mobility performance: Reflects reductions in travel time, delay, queue length, and congestion variability resulting from ITS deployment. Benefits are monetized using standardized value-of-time parameters consistent with federal guidance.
- Safety performance: Captures reductions in crash frequency and severity attributable to ITS technologies. Impacts are monetized using nationally recognized crash cost estimates by severity level.
- Environmental performance: Measures reductions in fuel consumption and vehicle emissions due to improved traffic operations. Benefits are monetized using fuel savings, emission factors, and the social cost of carbon parameters.

The evaluation incorporates two components for ITS costs:

- Operations and Maintenance Cost of Existing ITS Technologies: Represents the recurring expenditure required to sustain the operation of existing ITS infrastructure, including personnel, system maintenance, software support, communication services, and equipment repair or replacement.
- Design, Planning, Testing, and Deployment Cost of New ITS Technologies: Represents the annual investment required to develop and implement new ITS initiatives, including agency-funded design, planning, testing, and deployment projects with vendors, federal pass-through programs, and university research collaborations.

The benefit and cost values are standardized to constant evaluation-year dollars to ensure temporal comparability across reporting periods. The B/C ratio is also scaled to a value between 0 and 1, aligned with the entire rating system.

1.2 ITS Coverage in Existing Facility and Infrastructure

ITS coverage evaluates the extent to which ITS technologies are deployed across the eligible transportation network. Coverage is defined as the proportion of infrastructure assets in a jurisdiction equipped with operational ITS capabilities. This criterion measures system-wide deployment maturity and spatial penetration rather than isolated project installations.

Coverage assessment includes multiple infrastructure categories, which may include:

- ITS technologies and deployments, which are designed to improve safety and experience for vulnerable road users
- Signalized intersections equipped with advanced controllers, detection systems, adaptive signal control, or connected vehicle interfaces
- Freeway centerline miles covered by traffic management systems, dynamic message signs, CCTV surveillance, ramp metering, or traffic sensors
- ITS field devices, such as detectors, communications cabinets, controllers, and information displays
- Road weather information systems deployed for weather-responsive traffic management
- Connected vehicle roadside units (RSUs) providing V2X communication capability

The evaluation considers the total eligible infrastructure inventory within each category to ensure proportional measurement. Coverage reflects the degree to which ITS capabilities are embedded across the transportation network rather than limited to pilot deployments.

1.3 ITS Policy and Legislation Readiness

This criterion evaluates the policy and institutional environment supporting the planning, deployment, and operation of ITS. The assessment focuses on the existence of policy frameworks, legislative support, and the maturity of technology-related regulatory documentation that enable agencies to implement ITS programs.

The evaluation considers three components:

- **ITS Policy and Planning Documentation:** Reflects whether foundational strategic and institutional policy documents exist within the agency's governance framework. Examples include ITS strategic plans, regional ITS architectures, Vision Zero programs, and connected or automated vehicle task forces. The presence of these documents indicates formal policy commitment and institutional readiness for ITS deployment.
- **ITS Legislative Support:** Evaluates whether legislative actions support or constrain ITS implementation. Legislative instruments are reviewed to determine whether they enable, regulate, or restrict the deployment and operation of ITS technologies.
- **Maturity of ITS Technology Documentation:** Assesses the extent to which policies, regulations, or official documentation address the deployment and operational readiness of specific ITS technologies. Higher maturity levels indicate more developed regulatory frameworks and clearer authorization for technology implementation.

The three components collectively capture the policy and institutional conditions influencing the development and deployment of ITS within the evaluated jurisdiction.

1.4 ITS Integration in Funded and Planned Transportation Projects

This criterion evaluates the extent to which ITS is incorporated into future transportation investments through both funded initiatives and formal planning activities. Unlike indicators that measure deployment within existing infrastructure, this criterion focuses on forward-looking commitments that indicate whether ITS technologies are systematically integrated into upcoming transportation improvements.

The evaluation considers two aspects:

- **ITS Deployment through Funded Initiatives:** Reflects projects that have secured external funding or formal program support for ITS implementation. These projects represent confirmed investments and demonstrate institutional capability to advance ITS deployment. In particular, federally funded initiatives and large-scale funding awards indicate a higher level of competitiveness, institutional capacity, and commitment to implementing advanced ITS technologies.

- ITS Integration in Transportation Planning: Reflects ITS components identified within official transportation planning documents, capital improvement programs, or corridor development plans. These projects represent planned initiatives that indicate the agency's intention to incorporate ITS technologies into future infrastructure development.

Funded initiatives represent confirmed investments and therefore serve as the primary indicator within this criterion, reflecting demonstrated capability to secure resources and implement ITS technologies. Planning-based initiatives provide supplementary evidence of future integration intentions but carry a comparatively smaller influence in the overall evaluation.

1.5 ITS Facilities Capacity Evaluation

This criterion evaluates the institutional and operational capacity supporting the deployment, management, and advancement of ITS. The assessment focuses on the presence of organizational structures, technical resources, and supporting infrastructure that enable agencies to operate, maintain, test, and develop ITS technologies.

The evaluation considers whether agencies possess the necessary facilities and institutional resources to sustain ITS programs, support system maintenance, and facilitate technology development and innovation. These capacities reflect the agency's ability to manage ITS infrastructure, conduct system testing and evaluation, and maintain reliable system operations over time.

A higher level of facility capacity indicates a stronger institutional capability to support large-scale ITS deployment, long-term system operation, and continuous technological advancement.

2. Evaluation Data Assembling and Processing

2.1 ITS Deployment Cost and Benefit Database

The ITS Deployment Cost and Benefit Database is a structured repository that compiles documented cost information and quantified performance outcomes from Intelligent Transportation Systems deployments. The cost database contains multiple layers of cost information, including system-level deployment summaries, element-level cost ranges, and unit-level cost records. Cost entries generally include capital expenditures, operations and maintenance costs, service-life assumptions, dollar-year references, and technology classification fields. ITS service categories and subsystem classifications organize the data and are searchable by technology type, geographic context, and deployment characteristics.

The benefit database contains documented performance impacts from ITS evaluations. Each entry includes a structured description of the deployment context, evaluation methodology, and quantified outcomes across defined goal areas such as safety, mobility, efficiency, productivity, and customer satisfaction. Benefit entries are categorized by ITS application domain, performance goal area, and evaluation type. The database supports keyword search and categorical filtering, and entries can be exported for further analysis.

For this project, cost and benefit records are extracted and aligned with the ITS technology categories defined in the evaluation framework. Cost data are normalized to a consistent reference year to ensure comparability across deployments. Benefit metrics are standardized by performance domain and reorganized to match the mobility, safety, and environmental structure used in the annual evaluation criteria. The processed data are then incorporated into the incremental benefit–cost assessment model to support consistent economic evaluation across jurisdictions and reporting periods.

2.2 ITS Deployment Survey Data

The ITS Deployment Survey Data are derived from the ITS Deployment Tracking Survey (DTS). This recurring national data-collection effort documents the extent of ITS adoption across transportation agencies in the United States. The survey compiles structured responses from state and local transportation agencies regarding the presence, operational status, and functionality of deployed ITS technologies. Data is collected periodically across multiple survey cycles, enabling longitudinal analysis of deployment trends.

The survey database includes structured fields related to agency characteristics, system coverage, technology implementation status, functional capabilities, communications infrastructure, integration level, and operational maturity. Data is organized by technology domain and application area, allowing classification of deployments across roadways, transit, and freeways. Survey responses are typically coded as categorical or binary indicators, and in some cases include quantitative measures such as percentage coverage or deployment extent.

For this project, deployment survey data from all available survey years are aggregated to construct a longitudinal ITS deployment profile. Responses are standardized across survey cycles to ensure consistency in technology classification and coverage interpretation. Where survey structures evolve, variables are harmonized to maintain comparability. The processed data is used to quantify infrastructure coverage, assess deployment maturity levels, and track the temporal expansion of ITS capabilities across jurisdictions. The resulting indicators are incorporated into the ITS Scorecard coverage and penetration evaluation framework.

2.3 National Legislative Database

The National Legislative Database is constructed from publicly accessible legislative tracking resources to systematically monitor, search for, and document federal and state bills, statutes, and policy instruments related to Intelligent Transportation Systems (ITS) and emerging transportation technologies. Legislative records are collected annually and include enacted laws, proposed bills, amendments, appropriations measures, and regulatory provisions that may influence ITS planning, deployment, governance, and funding. The database captures bill identifiers, jurisdiction, status, enactment year, policy domain, and textual content for structured analysis.

This database supports longitudinal tracking of how the legislative environment evolves, enabling, accelerating, or constraining ITS deployment. Legislative monitoring focuses on identifying

statutory language that references transportation technology, digital infrastructure, connected and automated systems, mobility data platforms, and institutional governance structures related to intelligent transportation operations.

For this project, legislative documents are processed using structured text extraction and classification procedures. Each bill is first screened for ITS relevance from multiple perspectives, including subject-matter alignment, consistency with national ITS strategic objectives, integration with regional transportation planning priorities, and stakeholder impact. Legislation that meaningfully references ITS-related topics receives a baseline relevance designation.

2.4 National MPO Transportation Plan Database

The National MPO Transportation Plan Database consists of long-range transportation plans (LRTPs), metropolitan transportation plans (MTPs), Transportation Improvement Programs (TIPs), statewide transportation improvement programs (STIPs), capital improvement plans, corridor studies, and related planning documentation published by metropolitan planning organizations (MPOs), state departments of transportation, and regional agencies. These documents typically include structured project lists, programmed investment tables, financial plans, performance targets, implementation timelines, and modal investment breakdowns over multi-year planning horizons. Within these datasets, ITS-related projects are identified through project descriptions, technology components, system upgrades, communications infrastructure elements, data platforms, and operational improvement initiatives embedded within broader capital programs.

For scorecard purposes, this database supports the evaluation of ITS Penetration in Planned Asset Improvement/Expansion Projects by systematically extracting and classifying planned investments that incorporate ITS components. Project records are standardized across agencies to harmonize terminology, project types, cost allocations, funding sources, and implementation schedules. The processed dataset enables quantification of the proportion of planned transportation investments that explicitly integrate ITS technologies, modernization efforts, digital infrastructure upgrades, or smart mobility systems, thereby reflecting a forward-looking commitment to ITS integration within future capital improvement and expansion programs.

2.5 Agency-Based ITS Policy, Award, and Planning Documentation Scanning

Agency-based ITS policy, planning, and recognition documents are systematically collected through structured scanning of official agency websites and publicly available institutional repositories. The objective of this process is to identify authoritative materials that reflect an agency's strategic commitment to intelligent transportation systems, including ITS policy statements, strategic plans, technology deployment strategies, transportation improvement programs, research initiatives, and programmatic reports. In addition, documentation related to ITS testing facilities, pilot programs, operational guidelines, and institutional partnerships is reviewed to capture evidence of technological implementation and organizational readiness. These documents provide qualitative and quantitative indicators of how agencies incorporate ITS technologies into their operational frameworks and long-term transportation strategies.

The collected materials are then screened and categorized according to their relevance to the evaluation framework. Policy and legislative documentation support the legislative score, while planning documents and investment programs inform the assessment of planned ITS project penetration. Recognition materials, such as national ITS awards, innovation program acknowledgments, and agency-reported testing or demonstration facilities, provide additional indicators of technological leadership and institutional capability. By consolidating these materials across agencies, the documentation scanning process establishes a consistent evidence base that supports the rating methodology and ensures that each evaluated agency is assessed using verifiable, publicly accessible information.

3. Rating Methodology

3.1 ITS Criteria Grouping and Value Unification

Let E_i represent each evaluation criterion, where:

E_{BC} : Unified Annual Benefit/Cost Ratio Score (0-1)

E_{Cvg} : Unified ITS Coverage Percentage Score (0-1)

E_{Doc} : Unified ITS Policy and Legislation Documentation Score (0-1)

E_{Plan} : Unified Planned ITS Project Penetration Score (0-1)

$E_{Facilities}$: Unified Planned ITS R&D and Testing, Operational and Maintenance Facilities, Fleet, and Resources Score (0-1)

Each criterion will be normalized to a unified scale:

$$S = 100(w_1E_{BC} + w_2E_{Cvg} + w_3E_{Doc} + w_4E_{Plan} + w_5E_{Facilities})$$

Where w_i are the assigned weights for each evaluation criterion E_i tentatively assigned to 40% (B/C), 30% (coverage), 10% (Policy and legislative), 10% (ITS penetration in transportation plans), and 10% (facilities and resources).

3.2 Key ITS Sub-rating Generation Methods

3.2.1 Annual Benefit/Cost Ratio Score

The economic performance of ITS deployment is evaluated using an annual incremental benefit–cost framework. Rather than relying on a single lifecycle benefit–cost calculation, this methodology measures year-over-year performance improvement by comparing annual total ITS benefits to the total annual ITS investment, operations, and maintenance costs. The annual incremental benefit–cost ratio is defined as:

$$\Delta B/C(t) = \frac{\Delta B_{exist} + \Delta B_{new}}{\Delta C_{O\&M} + \Delta C_{new}}$$

Where t is the current year, ΔB_{exist} is total benefits that continuously received from existing ITS systems, ΔB_{new} is the benefits introduced by newly-deployed ITS systems, $\Delta C_{O\&M}$ is the current cost number of operations, maintenance, and repair, and ΔC_{new} is any ongoing spending on design, planning, testing, and deploying new ITS technologies through survey and getting budget and operating cost numbers from corresponding agency departments and units.

Here, B_t and C_t represent total monetized benefits and total annualized costs in year t , respectively. This structure ensures that the evaluation reflects performance improvement attributable to ITS expansion or operational enhancement within the evaluation period.

The existing or new annual benefit is further decomposed into mobility, safety, and environmental components:

$$\Delta B = \Delta B_{exist} + \Delta B_{new} = B^M + B^S + B^E$$

Each component is defined by using nationally recognized federal standards to ensure methodological consistency.

Mobility Benefit — USDOT Congestion Cost Savings

Mobility benefit is quantified as the total congestion cost savings resulting from travel time reductions enabled by deployed ITS technologies. First, we establish the travel time savings based on data from either before-and-after studies performed by targeted transportation agencies for their deployed technologies or using existing studies reported to the USDOT ITS benefit and cost database for the corresponding technologies. With such data, we estimate either percentage-based or time-value-based travel time savings, then convert these values to monetized values using local value-of-time data from transportation planning agencies, if available, or a federally recommended value.

The mobility benefit is derived from the reduction in excess travel time associated with improvements in the TTI. The incremental mobility benefit is expressed as:

$$B^M = \Delta T_{savings} \times D \times V \times VOT$$

where $\Delta T_{savings}$ is the total estimated travel time savings per day per vehicle (road user), D is the number of days when the corresponding technologies are operational, and V is the Annual Average Daily Traffic (AADT) on the deployed roadway infrastructure. Furthermore, these benefits will be calculated for each technology, including both existing technologies that are still operational and new technologies deployed in the evaluation year.

Safety Benefit — NSC 2023 Economic Cost by Injury Severity

Safety benefits are monetized using the National Safety Council (2023) Average Economic Cost by Injury Severity or Crash. Injury severity classifications follow Sections 2.3.4 through 2.3.6 of the Manual on Classification of Motor Vehicle Traffic Accidents (ANSI Standard D16.1-2007).

The safety benefit is calculated as:

$$B^S = \sum_k (\Delta N_k \times EC_k)$$

where ΔN_k represents the reduction in crashes of severity level k, and EC_k represents the average economic cost per crash for severity level k, based on 2023 national estimates. Severity categories include fatal injury, incapacitating injury, non-incapacitating injury, possible injury, and property-damage-only crashes.

The team will use existing before-and-after total reductions in crashes for the target ITS deployment. Traffic conflicts data may be converted into traffic crashes with an empirical formula reported in FHWA reports.

Environmental Benefit — EPA Standard Emission Valuation Framework

Environmental benefits are monetized using U.S. Environmental Protection Agency (EPA) emission factors and federally recognized social cost parameters. The environmental benefit consists of fuel savings and emission reduction savings:

$$B^E = (\Delta Fuel \times P_{fuel}) + \sum_j (\Delta E_j \times SCC_j)$$

where $\Delta Fuel$ represents a reduction in fuel consumption, P_{fuel} represents fuel price per unit, ΔE_j represents emission reductions of pollutant j, and SCC_j represents the social cost factor for pollutant j. In this project, pollutants are considered only CO₂, consistent with EPA modeling practices.

Operations and Maintenance Cost of Existing ITS Technologies

Existing ITS assets under agencies' operation incur recurring expenditures to keep them functional and reliable. In the annual incremental framework, these costs represent the baseline resource commitment required to sustain continuous operation and maintain the current level of service. The operations and maintenance cost is computed as the sum of labor, contracted services, utility and communication fees, routine preventive maintenance, corrective repairs, replacement parts, software support, and licensing costs for all operational ITS subsystems during the target year.

The annual operations and maintenance cost of existing ITS technologies can therefore be expressed as

$$\Delta C_{O\&M}(t) = C_{labor}(t) + C_{contract}(t) + C_{utility}(t) + C_{repair}(t) + C_{parts}(t) + C_{software}(t) + C_{license}(t)$$

where t denotes the target year; $C_{parts}(t)$

$C_{labor}(t)$ represents personnel costs for system operation and maintenance;

$C_{contract}(t)$ represents contracted service expenditures;

$C_{utility}(t)$ represents utility and communication service fees;

$C_{repair}(t)$ represents corrective repair expenditures;

$C_{parts}(t)$ represents replacement hardware and spare parts costs;

$C_{software}(t)$ represents software maintenance and technical support costs;

$C_{license}(t)$ represents annual software licensing fees.

Design, Planning, Testing, and Deployment Cost of New ITS Technologies

In addition to sustaining existing ITS operations, agencies often invest in developing and deploying new ITS technologies to expand system capabilities and improve transportation system performance. These investments typically involve multiple stages, including planning and concept development, system design and engineering, testing and validation, and field deployment. In the annual incremental evaluation framework, these expenditures represent the additional resource commitment required to introduce new ITS technologies and services during the target year.

The cost associated with new ITS technologies includes agency-funded ITS design, planning, and deployment projects conducted with vendors, as well as federally funded pass-through projects implemented through universities, research institutions, or other partner organizations. These investments may involve planning studies, system architecture design, pilot testing and validation, field installation and integration, and project management activities required to implement new ITS capabilities.

The annual cost associated with new ITS technology design and deployment can therefore be expressed as

$$\Delta C_{new}(t) = C_{plan}(t) + C_{design}(t) + C_{testing}(t) + C_{deploy}(t) + C_{manage}(t)$$

where t denotes the target year;

$C_{plan}(t)$ represents planning and concept development costs;

$C_{design}(t)$ represents system design and engineering expenditures;

$C_{testing}(t)$ represents testing, pilot operation, and validation costs;

$C_{deploy}(t)$ represents deployment and installation expenditures, including hardware procurement and infrastructure integration; and

$C_{manage}(t)$ represents project management and administrative costs associated with agency-funded ITS projects, vendor-supported implementations, and federally funded research or pilot programs.

Score Unification Method

The Benefit and Cost ratio can be any positive number. However, to unify this measure, we will use $B/C = 1$ as the unification standard, as it indicates that the cost of ITS O&M and new investment is fully covered by the benefits generated.

$$U_{BC} = \frac{r^2}{1 + r^2}$$

Where $r = B/C$. The sample unified rating from the B/C ratio can be found in the following table:

Table 1. Sample Unified Rating for B/C Ratio

B/C Ratio	Unified Rating
0.5	0.2
1	0.5
2	0.8
4	0.94

3.2.2 ITS Coverage Score

The ITS Coverage Score measures the extent to which ITS technologies are deployed across an agency's transportation infrastructure. The total number of infrastructure units, which is the total signalized intersections for arterial agencies or the total roadway miles for freeway agencies, is treated as a fixed reference base. Different ITS technologies may be implemented on different subsets of this infrastructure. Therefore, coverage unification is achieved by normalizing each technology deployment against the same total infrastructure base.

Let N^{Total} = the total number of ITS-applicable infrastructure units operated by the agency. For each ITS technology i , the normalized coverage percentage contribution is calculated as:

$$P_i = \frac{N_i}{N^{Total}}$$

where N_i is the number of infrastructure units on which technology i is deployed.

Then, the overall coverage score is calculated as the aggregated contribution of all evaluated technologies:

$$E_{avg} = \sum_{i=1}^n w_i P_i$$

where w_i is the weighted factor assigned to technology i , and n is the total number of ITS technology's evaluations.

Top Category of ITS Technology for Evaluation

To operate this framework across multiple ITS survey vintages, the Coverage Score should be built from survey-derived metric groups rather than from a single generic deployment count. In this project, the coverage dimension is expanded into a structured set of ITS deployment domains, and each survey item is normalized by response type before aggregation.

Specifically, each included survey item is first classified as: (1) a binary deployment or use condition, (2) a numeric deployment extent, or (3) an ordered categorical penetration or maturity response. Each response is then converted to a unified item score on a 0-1 scale, allowing heterogeneous survey formats to be combined within a single coverage framework.

For binary items, the default scoring logic is 1 for deployed, used, equipped, or active and 0 for not deployed or not used, while explicitly negative exclusive responses are polarity-adjusted so that higher readiness still receives the higher score.

For numeric items, including counts of equipped intersections, covered centerline miles, metered ramps, deployed roadside units, or technology-equipped vehicles, the item score is normalized using min-max scaling over the valid observed response range for that metric in the collected survey dataset. Where all valid responses are identical, any valid response receives full credit for that item.

For multiple-choice items, if the answer labels contain explicit percentage ranges, the selected response is converted using the stated percentage midpoint. If no percentage is provided, the item is normalized by ordered rank across valid response codes, with label-based polarity adjustments applied where the response wording clearly indicates increasing or decreasing deployment maturity.

Accordingly, the unified coverage score may be implemented as a weighted average of normalized item scores across all included coverage-related metrics, where each metric i contributes a unified item score P_i and an optional policy weight w_i . By default, equal weights may be used unless a policy-specific weighting structure is defined.

To improve comparability and auditability, the survey-derived coverage metrics should be grouped into the following ITS deployment domains, each with a corresponding unification approach:

Table 2. ITS Deployment Category for Coverage Evaluation

Coverage Domain	Example Survey-Derived Metrics	Recommended Unification / Rating Method	Suggested Weight Priority
Signal Management and Intersection Control	Central or coordinated signal control; adaptive signal control; queue detection; preemption; priority functions	Binary deployment scoring; normalized count or percent coverage; ordered maturity scoring	High: core arterial ITS foundation and direct operational coverage indicator
Traffic Monitoring and Data Collection	Real-time traffic monitoring coverage; detector deployment; probe readers; CCTV-supported monitoring; external data and open data feeds	Binary availability scoring; normalized network coverage; data maturity scoring	High: enables most downstream ITS operations, performance management, and active control
Vulnerable Road User Safety Applications	Pedestrian and cyclist detection at intersections; pedestrian hybrid beacons; passive pedestrian sensors; crossing conflict warning and related VRU-focused safety applications	Binary deployment scoring; normalized coverage at eligible locations; ordered safety-application maturity scoring	High: direct safety impact for pedestrians and cyclists, with strong alignment to Vision Zero and systemic safety goals
Traveler Information and User Services	DMS/CMS deployment; highway advisory radio; dissemination channels; real-time traveler information; trip planners; parking information services	Binary service presence; normalized asset coverage; ordered service capability scoring	Low: important user-facing outcome, but often available through navigation apps and services, maybe elevated for significant agency data exchange programs with ATIS providers
Active Traffic and Demand Management	Ramp metering; metered ramp coverage; lane control; managed lanes; reversible lanes; congestion pricing support systems	Binary operational deployment; normalized facility extent; ordered operational sophistication scoring	Low: important strategies with long history and adaptation by agencies, maybe elevated with strong benefit results.

Safety Enforcement and Incident Response	Automated enforcement; red-light enforcement; incident detection and verification; pedestrian warning; related safety deployment measures	Binary safety-system presence; normalized deployment extent; ordered safety capability scoring	High: closely tied to safety performance and incident-management readiness
Road Weather Information and Response	RWIS/ESS deployment; weather and road-condition data collection; adverse weather management strategies	Binary weather-system presence; normalized monitored-asset coverage; ordered weather-response readiness scoring	Medium: critical in weather-sensitive regions, but relevance varies by geography and exposure
Work Zone ITS and Queue Warning	Work zone ITS devices; smart work zone tools; queue warning systems; dynamic lane merge; temporary deployment technologies	Binary work-zone ITS presence; normalized temporary asset coverage; ordered operational maturity scoring	Medium: important for temporary risk management, but deployment is episodic rather than systemwide. May elevated to high for large-scale deployment or severe work zone congestion exists.
Transit and Fleet ITS Technology	AVL deployment; CAD/AVL support; real-time transit information; fare payment technologies; fleet technology penetration by service mode	Binary capability presence; normalized fleet penetration; ordered transit technology maturity scoring	Medium: highly important for transit agencies, but not equally central in all non-transit contexts, maybe elevated to High for transit-dependent communities
Connected, Automated, and Emerging Vehicle Technology	Connected vehicle deployment; roadside units; CV applications; AV testing; supporting communications and data-structure deployment	Staged readiness scoring; normalized unit deployment; binary deployment status scoring	Medium: strategic and forward-looking, but still uneven and less universal than core ITS deployments

ITS Program Planning and Operational Support	Documented plans; signal performance measurement; staff capability; training; certification; architecture; governance support	Binary institutional readiness; normalized staffing/resource extent; ordered organizational maturity scoring	Medium: essential enabling capacity, but more indirect than field deployment measures
--	---	--	---

Under this structure, the ITS Coverage Score reflects not only the extent of physical infrastructure coverage but also the degree to which ITS capabilities are deployed, scaled, and operationalized across the agency's network and supporting systems.

For missing or absent data, including blank responses, coded no-response values, and skip-logic values, the recommended default is to retain the record for transparency but exclude the item from the default denominator used in coverage aggregation unless a policy-specific imputation rule is explicitly adopted.

Score Unification Method

Since the P_i value could be any value from 0 to 1, and E_{cvg} may not directly show how important the growth of ITS coverage is among the total infrastructure. To make the overall coverage score consistent with a reasonable range between 0 and 1, a monotonic curve scale will be applied to unify the score:

$$U_{cvg} = \frac{E_{cvg}^{0.6}}{E_{cvg}^{0.6} + (1 - E_{cvg})^2}$$

The sample unified rating from the ITS overall coverage score can be found in the following table:

Table 3. Sample Unified Rating for ITS Overall Coverage Score

ITS Overall Coverage Score	Unified Rating
30%	0.50
50%	0.72
70%	0.90
90%	0.98

3.2.3 ITS Policy and Legislation Documentation Score

The ITS Policy and Legislation Documentation Score evaluate the extent to which an agency's institutional environment supports the planning, deployment, and operation of intelligent transportation systems. The assessment integrates three complementary aspects: (i) the existence of foundational ITS policy and planning documents, (ii) the level of legislative support related to ITS deployment, and (iii) the maturity of policy documentation associated with specific ITS technologies. These components collectively reflect both an agency's structural readiness and the policy maturity of ITS initiatives.

These three components are combined to produce the overall policy and legislation score:

$$L = L_{poli} + L_{legis}$$

ITS Policy and Planning Documentation

This section assesses whether key strategic and institutional policy documents are present within the agency's governance framework. The evaluation considers the presence of major planning or coordination mechanisms, including but not limited to:

- Vision Zero programs
- ITS Strategic Deployment Plans
- Autonomous Vehicle Task Forces
- Connected and Automated Vehicle Task Forces
- Regional ITS Architectures

Let D_j denote a binary indicator representing whether document j exists, where $D_j = 1$ if the document is available and $D_j = 0$ otherwise. If the number of m policy documents is evaluated and the policy document presence score is computed as how much documentation can be given over the total number of documents listed, as:

$$L_{poli} = \frac{\sum_{j=1}^m D_j}{m}$$

ITS Legislation Support

The Legislative Score quantifies the degree of policy support for ITS deployment using a five-level categorical framework. Each legislative document is coded into one of four categories:

$$L_i = \{-1, 0, 1, 2, 3\}$$

where $L_i = -1$ is assigned when legislation is restrictive, meaning it blocks or severely limits ITS implementation. A value of $L_i = 0$ indicates a neutral position in which the legislation addresses general road safety or transportation matters without explicit ITS relevance. A value of $L_i = 1$ to 3 denotes supportive legislation at different levels.

The overall legislative support score is then computed by the score for each evaluated legislation over how much legislation has been evaluated, as:

$$L_{legis} = \frac{\sum_{i=1}^n L_i^*}{n}$$

Score Unification Method

The ITS Policy and Legislation Documentation Score L can be any positive number. To make the overall score consistent with a reasonable range between 0 and 1, a monotonic curve scale will be applied to unify the score:

$$U_{Doc} = \frac{L^2}{1 + L^2}$$

The sample unified rating score can be found in the following table:

Table 5. Sample Unified Rating for ITS Policy and Legislation Documentation Score

ITS Policy and Legislation Documentation Weighted Counts	Unified Rating
0.5	0.2
1	0.5
2	0.8
4	0.94

3.2.4 Planned ITS Project Score

The Planned ITS Project Score evaluates the extent to which agencies are actively preparing for future ITS deployment through funded initiatives and formal transportation planning programs. Unlike indicators that measure existing operational infrastructure, this score focuses on forward-looking activities that demonstrate institutional commitment to ITS expansion. Two major sources are considered in this evaluation: federally recognized ITS-related grants and ITS projects identified in transportation planning documentation. Federally recognized grants reflect national-level competitiveness and innovation capacity, while planning documents indicate locally identified ITS initiatives and strategic investment priorities.

The overall Planned ITS Project Score combines these two components to represent the total level of planned ITS activity. Let S_{award} denote the score derived from federally recognized grants and S_{plan} denote the score derived from ITS projects identified in transportation planning documentation. The combined planned ITS project indicator is first calculated as:

$$P = S_{award} + S_{plan}$$

Federally-Recognized Grants Score

Federally recognized ITS grants provide strong evidence that an agency has successfully secured competitive national funding to deploy advanced transportation technologies. Programs considered in this evaluation include Advanced Transportation and Congestion Management Technologies Deployment (ATCMTD), Strengthening Mobility and Revolutionizing Transportation (SMART), Congestion Mitigation and Air Quality Improvement Program (CMAQ), Advanced Transportation Technologies and Innovative Mobility Deployment (ATTAIN), Rebuilding American Infrastructure with Sustainability and Equity (RAISE, formerly BUILD), Infrastructure for Rebuilding America (INFRA), Automated Driving Systems Demonstration Grants, and the Connected Vehicle Pilot Deployment Program.

Let N_g denote the number of federally recognized grants awarded to the agency and let F_g denote the total funding level associated with those grants. Agencies receiving at least one nationally recognized grant are considered to have achieved national-level recognition in ITS innovation and, therefore, receive a high baseline score. The award score is defined as:

$$S_{award} = \begin{cases} 0, & N_g = 0 \\ 0.8, & N_g \geq 1 \end{cases}$$

If the total funding level associated with the awarded grants exceeds a significant funding threshold θ , the score increases further to reflect the larger scale of the ITS investment. The funding adjustment is defined as:

$$S_{award} = 0.85 + 0.15(1 - \exp(-\frac{F_g - \theta}{\theta})), N_g \geq 1 \text{ and } F_g > \theta$$

ITS Projects Identified from Transportation Planning Documentation

In addition to federally recognized awards, agencies may also demonstrate planned ITS development through projects identified in transportation planning documentation. These documents may include metropolitan transportation plans, regional ITS architectures, corridor studies, transportation improvement programs, or other officially adopted planning reports. The evaluation focuses on planned ITS projects that have been formally identified for future implementation but may not yet have secured federal competitive funding.

Two measurable indicators are used to represent the scale of planned ITS initiatives: the number of planned ITS projects and the spatial coverage of planned ITS deployments. The first indicator captures the number of ITS projects listed in planning documents, while the second indicator reflects the total corridor mileage for planned ITS deployments. Either indicator may be used depending on the type of project documentation available.

Let N_p denote the number of planned ITS projects identified in planning documentation and let M_p denote the total mileage (in miles) of corridors associated with planned ITS deployments. A baseline score is assigned when the scale of planned ITS activity reaches a minimum meaningful

level, defined as either one to five planned ITS projects or more than ten miles of planned ITS deployment corridors. The baseline score is defined as

$$S_{plan} = \begin{cases} 0, & N_p = 0 \text{ or } M_p = 0 \\ 0.5, & 1 \leq N_p \leq 5 \text{ or } M_p > 10 \end{cases}$$

For larger planning programs, the score increases gradually as the scale of planned projects expands. This incremental increase is modeled using an exponential saturation function applied to the combined planning magnitude:

$$S_{plan} = 0.5 + 0.5 \times (1 - \exp(-\beta P_{N/M}))$$

Where

$$P_N = N_p - 5$$

$$P_M = M_p - 10$$

The exponential function allows the score to increase gradually as the scale of planned ITS projects grows, while keeping it bound between 0 and 1.

Score Unification Method

After the individual components for federally recognized ITS awards and ITS projects identified from transportation planning documentation are calculated, the final Planned ITS Project Score is obtained by combining the two components using a weighted aggregation. Because federally recognized ITS awards represent confirmed funding commitments and demonstrate national competitiveness in securing ITS deployment resources, they are given substantially greater importance in the final evaluation. Planning-based ITS initiatives represent future deployment intentions and therefore contribute a smaller proportion to the final score.

The unified Planned ITS Project Score is calculated as:

$$U_{plan} = w_{award} S_{award} + w_{plan} S_{plan}$$

The weighting parameters w_{award} and w_{plan} represent the relative contributions of funded initiatives and planned projects in the final evaluation. Because federally recognized ITS awards indicate confirmed investment and stronger implementation capability, they are assigned a dominant weight. In this study, the weights are defined as $w_{award}=0.9$ and $w_{plan}=0.1$, giving primary emphasis to award-based ITS deployment while allowing planning-based initiatives to provide supplementary contribution to the overall score. Since both S_{award} and S_{plan} are normalized within the interval $[0,1]$, and the weighting parameters satisfy $w_{award} + w_{plan} = 1$, the resulting unified score U_{plan} also remains within the interval $[0,1]$.

The sample unified rating from the ITS testing, operations, and maintenance facilities score can be found in the following table:

Table 6. Sample Unified Rating for Planned ITS Project Score

Planned ITS Project Weighted Count	Unified Rating
1	0.5
1.5	0.75
1.8	0.9
1.9	0.95

3.2.5 ITS Testing, Operations, and Maintenance Facilities Score

The ITS Testing, Operations, and Maintenance Facilities Score evaluates the physical and institutional infrastructure that supports the operation, testing, maintenance, and innovation of ITS technologies and deployments. This indicator measures an agency's operational capacity by quantifying the presence and scale of facilities that directly support ITS deployment and management. The evaluation considers several categories of operational and research infrastructure, including traffic operations centers; ITS operations and maintenance facilities and Fleets; ITS laboratories and research and development units; ITS research and resource centers or consortia; and ITS testbeds or pilot corridors for closure or open environment testing.

Each facility category is evaluated based on measurable indicators reflecting the scale of operational resources. These indicators may include the number of facilities, the number of associated operational units, such as maintenance fleets or laboratories, or the spatial coverage of testing infrastructure, such as the length of instrumented corridors or pilot deployment areas.

Let X_c denote the observed number for facility category c . Since these indicators may be reported in different units (e.g., counts or miles), the raw values cannot be directly compared across categories. Instead, they are aggregated to represent the overall operational capacity supporting ITS deployment. Let α_c denote the relative importance weight assigned to facility category c . The aggregated facility capacity indicator is defined as:

$$F = \sum_c \alpha_c X_c$$

Score Unification Method

Since the aggregated value F may take any nonnegative magnitude depending on the number and scale of facilities, a nonlinear transformation is applied to convert the capacity indicator into a standardized score between 0 and 1. The ITS Testing, Operations, and Maintenance Facilities Score is therefore calculated as:

$$U_{Facilities} = 1 - \exp(-0.7 \times F)$$

The sample unified rating from the ITS testing, operations, and maintenance facilities score can be found in the following table:

Table 7. Sample Unified Rating for ITS Testing, Operations, and Maintenance Facilities Score

ITS Policy and Legislation Weighted ITS Facility Count	Unified Rating
1	0.50
2	0.75
3	0.88
4	0.94

3.3 Final Rating Generation Methods

The overall ITS final score is computed as the weighted aggregation of the four standardized dimension scores derived from the evaluation framework. These dimensions represent economic efficiency, infrastructure deployment maturity, legislative readiness, and forward-looking planning integration. By design, each dimension score E_i has been normalized to a common 0 to 100 scale to ensure comparability before aggregation.

The composite score is defined as:

$$Final\ Score = \sum_{i=1}^5 w_i U_i \times 100$$

where α_i represents the assigned policy weight for evaluation dimension i, and E_i represents the standardized score for the four sub-dimensions. The weighting structure allows policy flexibility while maintaining mathematical consistency. Agencies may adopt equal weighting when all dimensions are considered equally important, or apply differentiated weights to reflect strategic priorities, such as emphasizing safety, modernization readiness, or capital planning integration.

The composite ITS Final Score is translated into categorical performance levels using a predefined threshold classification system. The purpose of this classification is to convert continuous quantitative scores into interpretable performance tiers that support cross-jurisdiction comparison, annual benchmarking, and policy communication.

The letter-grade classification thresholds are defined as follows:

- A+: ≥ 90
- A: 85 – 89

- **A–**: 80 – 84
- **B+**: 75 – 79
- **B**: 70 – 74
- **B–**: 65 – 69
- **C**: < 65

These thresholds are structured to differentiate high-performing ITS programs from moderate or emerging systems while preserving score sensitivity within each range. The upper tiers (A+, A, A–) indicate strong economic returns, high infrastructure penetration, supportive legislative environments, and consistent integration of ITS into planned capital programs. Mid-tier classifications reflect partial deployment maturity or uneven policy support, while lower classifications indicate limited deployment, weak institutional alignment, or insufficient integration into strategic planning.

This rating structure provides a standardized interpretation framework that enables longitudinal tracking of ITS development progress, transparent communication of system maturity, and objective comparison across agencies or states. By maintaining fixed threshold boundaries, annual evaluations remain consistent over time, ensuring methodological stability and reproducibility.

4. Application Architecture

4.1 ITS Scorecard Evaluation System Architecture

The ITS Scorecard application is implemented through a layered digital architecture that operationalizes the evaluation framework defined in Sections 1–3. The system integrates heterogeneous data sources, performs standardized processing and normalization, computes multi-dimensional performance ratings, and delivers interactive visual outputs through a web-based interface. Figure 1 illustrates the overall system architecture.

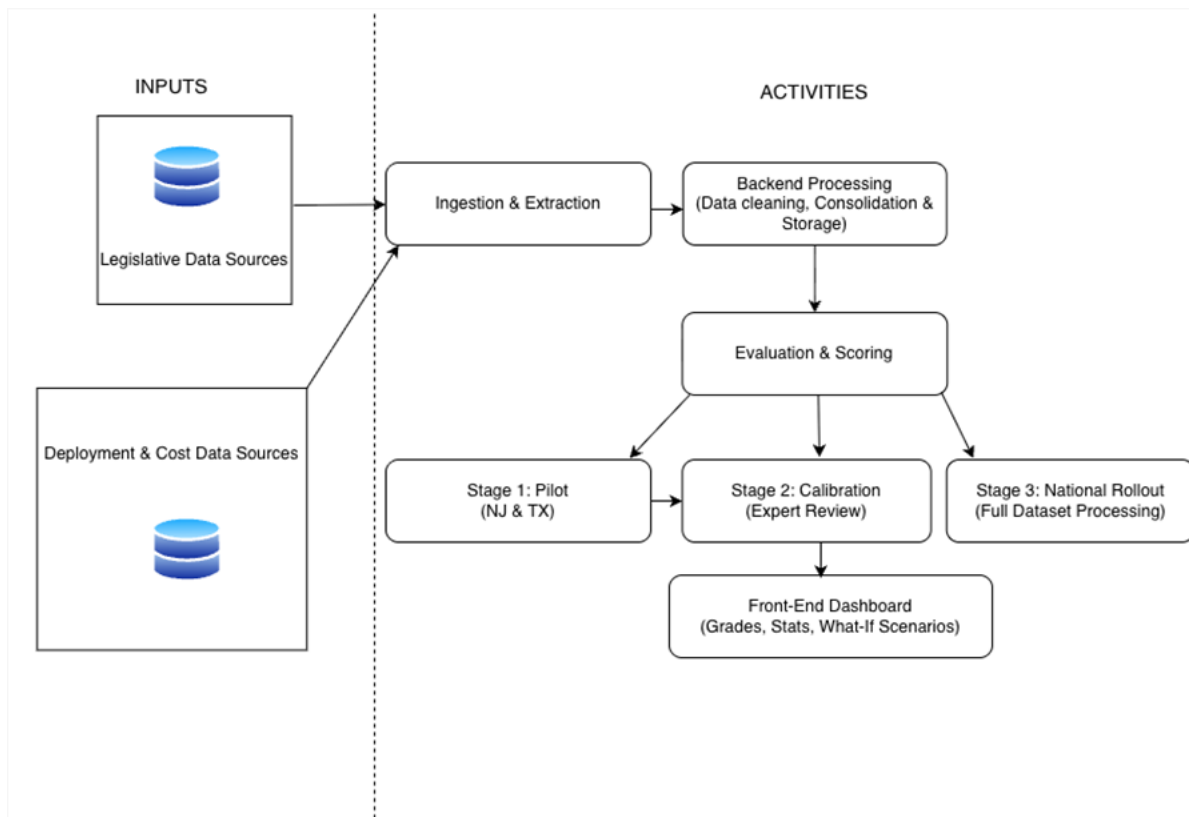


Figure 1. Layered Architecture of the ITS Scorecard Evaluation System

4.1.1 External Data Sources

The system aggregates data from multiple structured repositories to support comprehensive evaluation across the four core dimensions:

- ITS Deployment Cost and Benefit Database
- ITS Deployment Survey Data (multi-year longitudinal cycles)
- National Legislative Database
- National MPO Transportation Plan Database

- Agency-Based ITS Policy, Award, and Planning Documentation
- Expert panel and stakeholder input for weight calibration

These data sources provide the empirical foundation for economic evaluation, infrastructure coverage assessment, legislative scoring, and planning penetration analysis.

4.1.2 Central Data Repository (Supabase)

All datasets are stored in a centralized Supabase PostgreSQL environment that serves as the application's core data management layer. The repository contains:

- Raw ingestion tables preserving original data structures
- Standardized tables with harmonized variables
- Historical records for longitudinal tracking

This centralized structure ensures:

- Data integrity
- Version control
- Annual update capability
- Cross-jurisdictional comparability

Supabase also supports structured access control and API exposure for downstream application layers.

4.1.3 Data Processing Pipeline

The data processing pipeline transforms heterogeneous source data into unified evaluation inputs.

Key processes include:

- Survey harmonization across reporting years using canonical question mapping
- Validation and quality assurance against original survey dictionaries
- Handling of sub-question and non-response cases
- Standardization of variable formats (numeric vs categorical)
- Normalization of all evaluation metrics to a unified 0–100 scale
- Mapping of standardized variables to the four evaluation dimensions (E_i)

This structured pipeline ensures methodological consistency and reproducibility across annual evaluations.

4.1.4 Evaluation and Scoring Engine

The Evaluation and Scoring Engine implements the quantitative framework defined in Section 3.

The system computes five standardized subratings:

- E_{BC} : Unified Annual Benefit/Cost Ratio Score (0~1)

- E_{Cvg} : Unified ITS Coverage Percentage Score (0-1)
- E_{Doc} : Unified ITS Legislation Documentation Score (0-1)
- E_{Plan} : Unified Planned ITS Project Penetration Score (0-1)
- $E_{Facilities}$: Unified Planned ITS R&D and Testing, Operational and Maintenance Facilities, Fleet, and Resources Score (0-1)

Each dimension is normalized to a common scale and aggregated using a weighted composite formula to generate the final ITS Scorecard rating. The final numerical score is translated into categorical performance tiers (A+, A, A−, B+, B, B−, C) based on predefined thresholds.

All computed subratings and final scores are stored within the database to support transparency, auditability, and historical benchmarking.

4.1.5 Application Interface Layer

The final system layer provides interactive user access via a web-based dashboard. The interface includes:

- Map visualization of state-level ratings
- Detailed rating dashboards with subdimension breakdowns
- Query functionality for data transparency
- Planning and scenario simulation tools

The interface retrieves processed results via API endpoints connected to the centralized Supabase repository, ensuring real-time synchronization between backend computation and user-facing outputs.

The layered design of the ITS Scorecard architecture ensures scalability, modular updates, and methodological transparency. By separating data ingestion, processing, evaluation, and interface layers, the system maintains analytical integrity while supporting future expansion of evaluation dimensions or data inputs.

4.2 Key Functionalities

The ITS Scorecard application operationalizes the evaluation framework through structured analytical modules that compute, aggregate, and present performance metrics across the five evaluation dimensions E_i . The backend system performs standardized data ingestion, metric calculation, normalization, and score aggregation to support benchmarking, reporting, and scenario analysis. Each module corresponds directly to the evaluation methodologies described in Section 3.2 and processes the relevant datasets to produce normalized component scores used by the user interface dashboards.

4.2.1 Economic Performance Evaluation Module

Implements: Section 3.2.1 (Incremental B/C Framework)

This module evaluates the economic effectiveness of ITS deployment using the incremental benefit–cost methodology.

The module performs the following functions:

- Process the ITS deployment cost and benefit datasets
- Calculate annual incremental benefits and costs associated with ITS deployment
- Decompose benefits into mobility, safety, and environmental categories
- Aggregate monetized benefits across the three domains
- Generate the annual ITS economic performance score
- Normalize the result to the standardized scoring scale

4.2.2 Infrastructure Coverage Module

Implements: Section 3.2.2

This module evaluates the level of ITS deployment across transportation infrastructure assets. The module performs the following functions:

- Process deployment datasets for ITS technologies across roadway infrastructure
- Calculate deployment coverage for each technology category
- Apply technology-specific weighting factors
- Aggregate weighted coverage values across infrastructure elements
- Generate a unified ITS infrastructure coverage score
- Normalize the final score to the standardized evaluation scale

4.2.3 Policy and Legislation Documentation Scoring Module

Implements: Section 3.2.3

This module evaluates the policy and institutional environment supporting ITS deployment. The module performs the following functions:

- Process policy, legislative, and regulatory documentation datasets
- Identify ITS-related policy and planning documents
- Classify legislative instruments based on their policy impact
- Evaluate technology policy maturity using regulatory documentation
- Aggregate scores across policy documentation, legislative support, and technology maturity components
- Generate the unified policy readiness score for the evaluated jurisdiction

4.2.4 Planned ITS Penetration Module

Implements: Section 3.2.4

This module evaluates future ITS deployment readiness based on funded initiatives and planned transportation investments. The module performs the following functions:

- Identify ITS projects supported by external funding sources
- Extract ITS initiatives from transportation planning and capital improvement documents
- Calculate the score associated with funded ITS deployment initiatives
- Calculate the score associated with planned ITS deployment activities
- Combine the two components using the weighted aggregation method defined in the evaluation framework
- Generate the final Planned ITS Project Score

4.2.5 ITS Facilities Capacity Evaluation Module

Implements: Section 3.2.5

This module evaluates the institutional and operational capacity supporting ITS deployment and system management. The module performs the following functions:

- Process datasets describing ITS operational infrastructure and institutional resources
- Identify facilities and organizational resources that support ITS operations, maintenance, testing, and technology development
- Evaluate facility capacity indicators associated with ITS deployment support
- Aggregate facility capacity indicators across the evaluated jurisdiction
- Generate the unified ITS Facilities Capacity Score
- Normalize the final score to the standardized evaluation scale

4.2.6 Final Score Aggregation Module

Implements: Section 3.3

This module integrates the results from all evaluation dimensions and generates the final ITS Scorecard rating for the evaluated jurisdiction. The module serves as the system's central aggregation engine, combining normalized component scores to produce the overall performance assessment.

The module performs the following functions:

- Retrieve normalized scores from the five evaluation modules corresponding to the evaluation dimensions
- Apply predefined weighting parameters associated with each evaluation dimension
- Aggregate the weighted component scores to generate the final ITS Scorecard rating

- Assign a rating category or performance grade based on the predefined score classification thresholds
- Store the final score, component scores, and associated metadata within the system database
- Maintain historical records of annual evaluations to support longitudinal performance tracking and trend analysis
- Provide aggregated results to the visualization modules used by the dashboard interface

4.2.7 Forecasting & Scenario Simulation Module

This module supports forward-looking analysis by enabling users to simulate potential changes in ITS deployment conditions and evaluate their impact on the overall ITS Scorecard rating. The module extends the evaluation framework beyond retrospective assessment by supporting planning and decision-making through scenario analysis.

The module performs the following functions:

- Allow users to adjust key ITS development parameters, including:
 - projected ITS investment levels
 - anticipated expansion of ITS infrastructure coverage
 - changes in policy or legislative support conditions
 - planned ITS deployment initiatives or infrastructure programs
- Recalculate the affected evaluation dimensions based on the modified input parameters
- Generate projected component scores for each evaluation dimension
- Compute the resulting projected ITS Scorecard rating under the simulated scenario
- Compare projected outcomes with the baseline evaluation results
- Provide comparative outputs for visualization in forecasting dashboards and analytical reports